

Single choice Question with one option correct	Q.01 to Q.15
Multiple choice Question with one or more than one may be correct	Q.16 to Q.35
Numerical Based Question (no negative marking)	Q.36 to Q.55

DPP Syllabus : Quantum number, Periodic table, Chemical bonding, Coordination compound, s-block, Electrochemistry, Surface chemistry, Solution & colligative properties.

- The cell potential for the unbalanced chemical reaction :

$$\text{Hg}_2^{2+} + \text{NO}_3^- + \text{H}_3\text{O}^+ \rightarrow \text{Hg}_2^{2+} + \text{HNO}_2 + \text{H}_2\text{O}$$

under standard state conditions is $E^\circ_{\text{cell}} = 0.02 \text{ V}$

At what pH will the cell potential be zero if the concentration of other components are equal to one ?

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{2}{9}$
- The ligand $\text{N}(\text{CH}_2\text{CH}_2\text{NH}_2)_3$ is :

(A) tridentate (B) tetradentate (C) didentate (D) pentadentate
- The correct electronic configuration for $[\text{CrF}_6]^{4-}$ will be :

(A) $t_{2g}^4 e_g^0$ (B) $e_g^2 t_{2g}^2$ (C) $t_{2g}^3 e_g^1$ (D) $e_g^4 t_{2g}^0$
- The conductivity of a solution which is 0.2M in AgNO_3 and 0.1M in $\text{Ba}(\text{NO}_3)_2$ is :

$[\lambda^\circ(\text{Ag}^+) = 6 \times 10^{-3} \text{ Sm}^2\text{mol}^{-1}, \lambda^\circ(\text{Ba}^{2+}) = 13 \times 10^{-3} \text{ Sm}^2 \text{mol}^{-1} \text{ and } \lambda^\circ(\text{NO}_3^-) = 7 \times 10^{-3} \text{ Sm}^2\text{mol}^{-1}]$

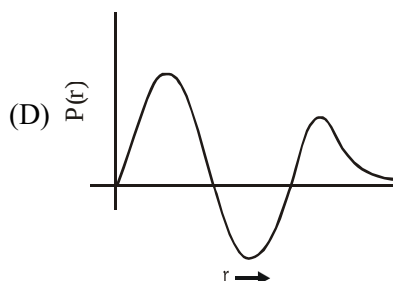
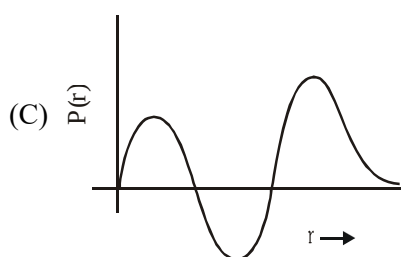
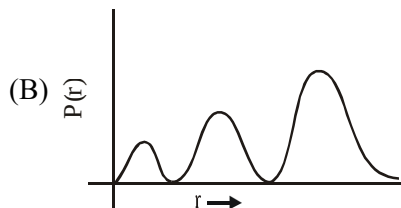
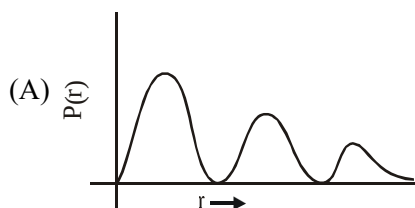
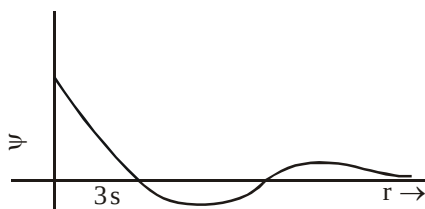
(A) 5.3 Sm^{-1} (B) 4.3 Sm^{-1} (C) 3.3 Sm^{-1} (D) 2.3 Sm^{-1}
- At 70 K, the adsorption of N_2 gas at iron surface obeys freundlich adsorption isotherm. The experimental data is :

P(bar)	4	25	64
$\frac{x}{m}$	0.2	0.5	0.8

Where x is the mass of N_2 in gram, adsorbed per gram of iron at P bar pressure. The mass of N_2 gas adsorbed by 5g of Fe at 36 bar and 70 K is :

- (A) 0.6 g (B) 3g
(C) 0.8 g (D) Cannot be found from given data

6. The wave function (ψ) versus radial distance (r) curve for certain orbital is given. Predict the shape of $\psi^2 \cdot 4\pi r^2$ (radial probability distribution function) versus r graph.



7. E_n = total energy
 λ_n = angular momentum in n^{th} orbit
 K_n = K.E., V_n = P.E.
 T_n = time period, r_n = radius of n^{th} orbit

	Column - I		Column - II
(I)	$E_n^{-y} \propto r_n$ then y , for a given Z , is	(p)	$\frac{1}{2}$
(II)	$\ell_n \propto n^x$ then x is	(q)	-2
(III)	$\frac{E_n}{V_n}$ is	(r)	-3
(IV)	$T_n \propto \frac{Z^t}{n^m}$, t is	(s)	1

Select the correct option :

- (A) (I) $\rightarrow$ s; (II) $\rightarrow$ s; (III) $\rightarrow$ p; (IV) $\rightarrow$ q (B) (I) $\rightarrow$ p; (II) $\rightarrow$ q; (III) $\rightarrow$ r; (IV) $\rightarrow$ q
 (C) (I) $\rightarrow$ s; (II) $\rightarrow$ s; (III) $\rightarrow$ q; (IV) $\rightarrow$ p (D) (I) $\rightarrow$ p; (II) $\rightarrow$ s; (III) $\rightarrow$ p; (IV) $\rightarrow$ q

8.	Column-I	Column-II
(i)	XeO_2F_4	(p) Central atom is sp^3 hybridized with no lone pair on it.
(ii)	SO_2Cl_2	(q) Central atom is sp^2 hybridized.
(iii)	COCl_2	(r) Central atom involves $\text{d}_{x^2-y^2}$ orbital in hybridization.
(iv)	HClO_3	(s) Contain central atom in oxidation state, two less than the highest.

Select the correct option :

(A) (i) $\rightarrow$ r ; (ii) $\rightarrow$ p ; (iii) $\rightarrow$ q ; (iv) $\rightarrow$ s (B) (i) $\rightarrow$ p ; (ii) $\rightarrow$ q ; (iii) $\rightarrow$ r ; (iv) $\rightarrow$ s

(C) (i) $\rightarrow$ q ; (ii) $\rightarrow$ r ; (iii) $\rightarrow$ p ; (iv) $\rightarrow$ s (D) (i) $\rightarrow$ s ; (ii) $\rightarrow$ q ; (iii) $\rightarrow$ r ; (iv) $\rightarrow$ p

9. **Statement-1** : H_2S is less acidic than H_2Te .

Statement-2 : As the size of Te is more than S, the bond Te-H is weaker than S-H .

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

10. **Statement-1** : Aluminium and zinc metal evolve H_2 gas from NaOH solution.

Statement-2 : Several non-metals such as P, S, Cl, etc. yield a hydride instead of H_2 gas from NaOH .

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

11. At 300 K, the vapour pressure of an ideal solution containing 3 mole of A and 2 mole of B is 600 torr. At the same temperature, if 1.5 mole of A & 0.5 mole of C(non-volatile) are added to this solution the vapour pressure of solution increases by 30 torr. What is the value of P_B^0 in torr?

(A) 940

(B) 405

(C) 90

(D) 900

12. Which of the following liquid pairs shows a positive deviation from Raoult's law ?

(A) Acetone – chloroform

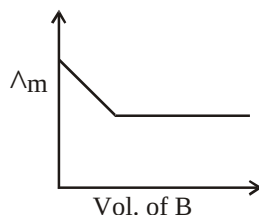
(B) Benzene – methanol

(C) Water – nitric acid

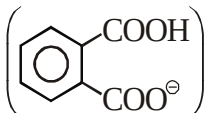
(D) Water – hydrochloric acid

13. The effect of temperature and pressure on adsorption is/are :
- (A) The amount of gas adsorbed per gram of adsorbent does not vary linearly with pressure at constant temperature if the pressure is either very low or very high
- (B) Chemisorption decreases with increase of temperature at high temperature
- (C) There is no effect of pressure on physisorption if temperature is kept constant
- (D) Higher the critical temperature of a gas, less is its physical adsorption
14. For the coagulation of 125 ml of $\text{Fe}(\text{OH})_3$ sol in 2 hrs, 2ml of 1.24 M Na_2SO_4 electrolyte is required. What is the coagulating value of Na_2SO_4 electrolyte ?
- (A) 19.84 (B) 125 (C) 1.24 (D) 2.48
15. Which of the following have an incorrect order of ionization energy :
- (A) $\text{Pb}(\text{IE}) > \text{Sn}(\text{IE})$ (B) $\text{Na}^+(\text{IE}) > \text{Mg}^+(\text{IE})$
- (C) $\text{Li}^+(\text{IE}) < \text{O}^+(\text{IE})$ (D) $\text{Be}^+(\text{IE}) < \text{C}^+(\text{IE})$
16. Sodium carbonate reacts with :
- (A) $\text{MgCl}_2(\text{aq.})$ (B) $\text{Ca}(\text{HCO}_3)_2$ (C) H_2SO_4 (D) dry CO_2
17. A gas 'X' undergoes physisorption on charcoal. If the magnitude of enthalpy change and entropy change were observed to be 20 kJ/mol and 75 J/mol-K, then the temperature at which adsorption takes place spontaneously is/are :
- (A) $T = 280 \text{ K}$ (B) $T = 300 \text{ K}$
- (C) $T = 250 \text{ K}$ (D) At any temperature adsorption will be spontaneous
18. Which of the following statements is/are correct?
- (A) Higher is the gold number of a lyophilic sol, lesser is its coagulating power.
- (B) An electrolyte can coagulate both positive sol and negative sol.
- (C) Impurities present in a colloid makes it more stable.
- (D) Dialysis is a process to remove impurities of ions and molecules from a colloid.
19. At 298K the standard free energy of formation of $\text{H}_2\text{O}(\lambda)$ is -257.20 kJ/mole while that of its ionisation into H^+ ion and hydroxyl ions is 80.35 kJ/mole , then (take $F = 96500 \text{ C}$)
- $\text{H}_2(\text{g}, 1 \text{ bar}) \mid \text{H}^+(1\text{M}) \parallel \text{OH}^-(1\text{M}) \mid \text{O}_2(\text{g}, 1\text{bar})$
- (A) $E_{\text{cell}} = 0.40 \text{ V}$ (B) $E_{\text{cell}} = 0.50 \text{ V}$
- (C) $\Delta G_{\text{cell}}^\circ = -96.5 \text{ kJ/mol of H}_2$ (D) $\Delta G_{\text{cell}}^\circ = -9.65 \text{ kJ/mol of H}_2$

20. Which of the following (B) reagents when added to (A) reagent drop by drop, result in the given variation of molar conductance?

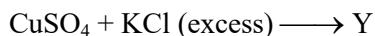
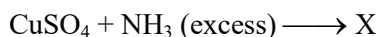


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|---|---|
| <p>A B</p> <p>(A) HCl + NH₃</p> <p>(C) NaOH + HCOOH</p> | <p>A B</p> <p>(B) NaCl + AgNO₃</p> <p>(D) BaCl₂ + Ag₂SO₄</p> |
|---|---|
21. On electrolysis, in which of the following, O₂ would be liberated at the anode?
- (A) dilute H₂SO₄ with Pt electrodes
- (B) aqueous AgNO₃ solution with Pt electrodes
- (C) dilute H₂SO₄ with Cu electrodes
- (D) aqueous NaOH solution with Fe cathode & Pt anode
22. In which of the following pairs, the number of lone pair(s) over central atom and molecular geometry is/are different ?
- (A) TeCl₄, PCl₄⁺ (B) ClF₃, PCl₃ (C) SiCl₄, BF₄⁻ (D) BrF₅, XeF₅⁻
23. Which of the following pairs of coordination compound show same type of isomerism?
- (A) [Co(NH₃)₅Br] SO₄ and [Ni(dmg)₂]
- (B) [Pt(NH₃)₂Cl₂] and [Co(en)₂BrCl] Cl
- (C) [Co(NH₃)₆][Cr(CN)₆] and K[PtCl₃(C₂H₄)]
- (D) [Ni(en)₃]²⁺ and [Co(ox)₃]³⁻

24. In which of the following pairs of elements the electronegativity of the first element is more than that of second element ?
- (A) Al, Ga (B) Li, K (C) Cl, S (D) F, Cl
25. Which among the following is stronger lewis bases than $\text{HN}(\text{SiH}_3)_2$?
- (A) NH_3 (B) $\text{H}_2\text{N}(\text{CH}_3)$ (C) $\text{H}_2\text{N}(\text{SiH}_3)$ (D) $\text{HN}(\text{CH}_3)(\text{SiH}_3)$
26. In which of following silicate(s), the number of corners shared per tetrahedron is ≥ 2 ?
- (A) Four membered cyclic silicate (B) Pyrosilicate
(C) Chain silicate (D) 2-D silicate
27. Which of the following is/are correct?
- (A) van der Waals radius of iodine is more than its covalent radius
(B) All isoelectronic ions belong to the same period of periodic table
(C) First ionisation energy of Be is more than that of C
(D) Electron affinity of N as well as noble gases is negative.
28. Which of the following show intramolecular hydrogen bonding?
- (A) salicylaldehyde (B) o-nitrotoluene
(C) p-nitrophenol (D) hydrogenphthalate ion 

Comprehension (One or more than one correct options)

Answer the questions based on the following reactions :

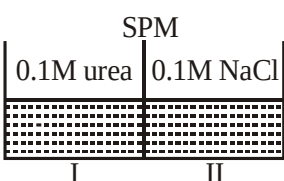


29. Which of the following is/are correct ?
- (A) color of X = Deep blue (B) color of X = red
(C) hybridization of Cu^{2+} in X = sp^3 (D) hybridization of Cu^{2+} in X = dsp^2

30. To Y, excess KCN is added due to which :
- (A) A highly stable complex $[\text{Cu}(\text{CN})_4]^{2-}$ is formed which remains in solution even after long time
- (B) A gas is produced
- (C) A highly stable complex $[\text{Cu}(\text{CN})_4]^{3-}$ is formed which remains in solution even after long time
- (D) The products are diamagnetic
31. Among the following compounds of sulphur, which contain sulphur-sulphur bond?
- (A) $\text{H}_2\text{S}_2\text{O}_5$ (B) $\text{H}_2\text{S}_2\text{O}_3$ (C) $\text{Na}_2\text{S}_4\text{O}_6$ (D) CaS_2O_7
32. The vapour pressure of two pure liquids X and Y which form an ideal solution are 200 and 300 mm of Hg respectively at 300 K. A liquid solution of 'X' and 'Y' in which the mole fraction of 'X' is 0.5 is contained in a cylinder closed by a piston on which pressure can be varied. Which of the following is/are incorrect ?
- (A) Only if pressure above solution is less than 200 torr, vapour phase can exist.
- (B) Only if pressure above solution is less than 300 torr, vapour phase can exist.
- (C) Only if pressure above solution is more than 200 torr, liquid phase exist.
- (D) Only if pressure above solution is less than or equal to 250 torr, vapour phase can exist.
33. Element 'A' has principal quantum number 2 for last electron and it has 3 electron in valence shell and element 'B' has principal quantum number 3 for last electron and it has seven electrons in the valence shell. The atomic number of an element X is equal to sum of atomic numbers of A and B. Which of the following statement is/are correct for X ?
- (A) M shell contain 10 electrons
- (B) Total number of electrons with $l = 1$ are 12
- (C) Total number of electrons with $m = 0$ may be 14
- (D) It can show back bonding in which d orbitals are involved.

34. An ideal binary liquid solution of A and B has a vapour pressure of 900 mm of Hg. It is distilled at constant temperature till $\frac{2}{3}$ of the original amount is distilled. If the mole fraction of 'A' in the residue and mole fraction of B in the condensate is 0.3 and 0.4 respectively and vapour pressure of residue is 860 mm of Hg, then which of the following is/are correct options :

- (A) The initial mixture taken should be an equimolar mixture of A and B.
 (B) The first vapors formed will have more moles of A as compared to moles of B.
 (C) $P_A^\circ = 1000$ torr.
 (D) The vapor pressure above the condensate is 920 torr at same temperature.

35. 

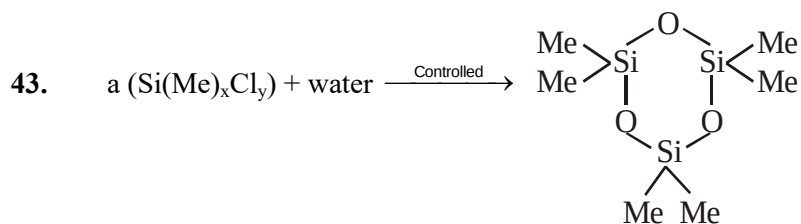
0.1 M urea and 0.1 M NaCl aqueous solutions of same volume are separated by semi-permeable membrane. Identify the true correct statement(s).

- (A) No water flow across the membrane
 (B) Water will net flow from I to II
 (C) Water will net flow from II to I
 (D) Both solutions will be isotonic in the end
36. The Henry's law constant at T K for dissolution of a gas in aqueous medium is 3×10^4 atm. At what partial pressure of the gas (in atm) at T K, the molality of gas in aqueous solution will be $\frac{5}{900}$?

37. How many of the following compound can be optically active?

- (a) $[\text{Co}(\text{en})_2\text{ClBr}]$
 (b) $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$
 (c) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]^+$
 (d) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
 (e) $[\text{Co}(\text{en})_3]^{3+}$
 (f) $[\text{Ni}(\text{dmg})_2]$
 (g) $[\text{Pb}(\text{EDTA})]^{2-}$
 (h) $[\text{Cr}(\text{ox})_2(\text{dmg})]^{2-}$
 (i) $[\text{Cr}(\text{dmg})_3]$

38. The sum of oxidation state of Co and Cr in the following complex is :
- $$[\text{Co}(\text{NH}_3)_6] [\text{Cr}(\text{NH}_3)_2\text{Cl}_4]_3$$
39. How many of the following species use d_{z^2} orbital in their hybridization?
- $[\text{Ni}(\text{CN})_4]^{2-}$
 - $[\text{CoBr}_4]^{2-}$
 - NH_4Cl
 - XeF_4
 - $[\text{Cr}(\text{NH}_3)_6]^{2+}$
 - $[\text{Cu}(\text{NH}_3)_4]^{2+}$
 - $[\text{FeF}_6]^{3-}$
40. Find total number of N-Cr-O angles in tris(glycinato)chromium (III).
41. Find the value of $x + y$ where x and y indicate number of H atoms in $\text{H}_x\text{Fe}(\text{CO})_4$ and $\text{H}_y\text{Re}(\text{CO})_5$ respectively. (EAN rule is followed).
42. How many of the following relations is/are correct ?
- $\text{Mg} > \text{Al}$ (electropositive character)
 - $\text{Al} > \text{Ga}$ (electronegativity)
 - $\text{N}^+ > \text{N}$ (Ionization energy)
 - $\text{Se} > \text{S}$ (magnitude of $\Delta_{\text{eg}}H$)
 - $\text{He}^+ > \text{H}$ (Atomic size)
 - $\text{O}_2^{2-} < \text{O}_2$ (paramagnetic nature)
 - $\text{C}_2 > \text{N}_2$ (Number of π bonds)
 - $\text{F}_2 > \text{B}_2$ (bond order)
 - $\text{B}_2 > \text{F}_2$ (bond length)
 - $\text{NO}_2^- > \text{NO}_3^-$ (bond angle)



What is the value of $a + x + y$?

44. How many of the following statements is/are correct?

- (i) The discharging of lead storage battery constitutes a galvanic cell.
- (ii) During charging of lead storage battery, anode is negatively charged.
- (iii) The mass lost at anode is equal to mass gained at cathode in any galvanic cell.
- (iv) E_{cell} is intensive quantity and is independent of temperature.
- (v) The overall reaction in an electrolytic cell need not always look like a redox reaction.
- (vi) If E_{cell}^0 of a cell reaction is negative, then the reaction is non-spontaneous under all conditions.

45. Pure water is saturated with pure solid AgCl. A silver rod is placed in the solution and the potential is measured against normal calomel electrode at 25°C . This experiment is then repeated with a saturated solution of AgI. If the difference in potential in the two cases is 0.177V, the ratio of solubility product (K_{sp}) of AgCl and AgI at the temperature of the experiment is 10^x . Determine the value of x .

46. During the preparation of $\text{H}_2\text{S}_2\text{O}_8$ (peroxydisulphuric acid) using H_2SO_4 electrolytically, O_2 gas also releases at anode as by product. When 8.96 L of H_2 releases at cathode and 2.24 L O_2 at anode at NTP, the weight of $\text{H}_2\text{S}_2\text{O}_8$ produced in gram is : (Round it off to nearest whole number)

47. 1 g of charcoal adsorbs 100 mL of 0.5 M CH_3COOH to form mono layer and thereby the molarity of CH_3COOH reduces to 0.49 M. Calculate the surface area of the charcoal adsorbed by each molecule of CH_3COOH . Surface area of charcoal = $3.01 \times 10^2 \text{ m}^2/\text{gm}$. Give your answer by multiplying with 10^{19} .

48. Find the total number of alpha particles and neutrons emitted when ^{12}C is bombarded on $^{139}_{57}\text{La}$ to produce $^{147}_{63}\text{Eu}$. (Assume no other particles are produced)
49. Calculate the maximum total number of electrons in Gallium, with spin $+\frac{1}{2}$ and which occupy orbitals having minimum one radial node.
50. ^{58}Zn decays by two paths. By path (I), it produce daughter nuclide 'X' along with a positron and proton. By path (II) it produce daughter nuclide 'Y' along with a positron. Find the sum of neutrons in X and Y.
51. 20 g of naphthoic acid ($\text{C}_{11}\text{H}_8\text{O}_2$) is dissolved in 50 g of benzene ($K_f = 1.72 \text{ K kg mol}^{-1}$). If the van't Hoff factor (i) of naphthoic acid is 0.5 then the value of depression in freezing point (in K) will be :
52. The by-product of Solvay process reacts with Na_2CO_3 to form a compound X, which on heating decomposes to give Y. Y is absorbed by KO_2 . The number of atoms per molecule of Y is
53. How many of the following will turn moist red litmus blue and finally white?

 Li_2O , KO_3 , RbO_2 , Cs_2O_2 , BeO , MgO , BaO_2 , SrO
54. Among the following, the number of compounds that would require electrolysis process to get their respective metal from their metal compound is _____.

 NaCl , Cr_2O_3 , MgCl_2 , Al_2O_3 , CaCl_2 , Fe_2O_3
55. The number of bicarbonates that do not exist in solid form among the following is _____.

 LiHCO_3 , NaHCO_3 , $\text{Ca}(\text{HCO}_3)_2$, KHCO_3 , NH_4HCO_3 , $\text{Ba}(\text{HCO}_3)_2$, $\text{Mg}(\text{HCO}_3)_2$

ANSWER-KEY DPP # 05 (MAIN + ADVANCED)

1.	(D)	2.	(B)	3.	(C)	4.	(A)	5.	(B)
6.	(B)	7.	(A)	8.	(A)	9.	(B)	10.	(C)
11.	(C)	12.	(B)	13.	(B)	14.	(A)	15.	(C)
16.	(ABC)	17.	(AC)	18.	(ABD)	19.	(BC)	20.	(AC)
21.	(ABD)	22.	(ABD)	23.	(BD)	24.	(BCD)	25.	(ABCD)
26.	(ACD)	27.	(AD)	28.	(AD)	29.	(AD)	30.	(BCD)
31.	(ABC)	32.	(ABC)	33.	(ABD)	34.	(ABCD)	35.	(BD)
36.	(0.3 atm)	37.	(6)	38.	(6)	39.	3 (d, e, g)	40.	(9)
41.	(3)	42.	(4)	43.	(7)	44.	2((i), (v))	45.	(6)
46.	(39)	47.	(5)	48.	(4)	49.	(7)	50.	(58)
51.	(2)	52.	(3)	53.	(4)	54.	(4)	55.	(4)